

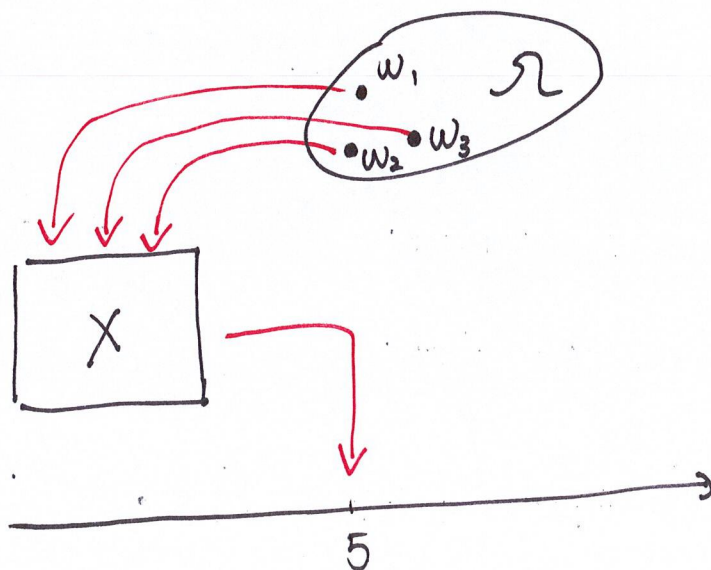
Discrete random variables

▮ $P(A)$ is the probability that event A happens.

▮ $P(X)$ is meaningless since X is a function.

$$\nabla P(X = x) = P(\{\omega \in \Omega \mid X(\omega) = x\})$$

▮ Eg.



$$P(X=5) = P(\{\omega_1, \omega_2, \omega_3\})$$

Def: Probability mass function (PMF) of a discrete RV is

$$P_X(x) = P(X=x)$$

NOTE : • A RV is denoted by an **upper** case letter (e.g. X, Y, Z).

• A value is denoted by a **lower** case letter (e.g. x, y, z)

Properties of PMF's :

• $P_X(x) \in [0, 1]$ for all $x \in (-\infty, \infty)$.

• $\sum_{\forall x \in \text{supp}(X)} P_X(x) = 1$

Ex. Keep tossing a fair coin until the first head is observed.

Let X be the # of tosses needed.

Find $P_X(x)$ and $P(X \leq 3)$.

$$\begin{aligned} \bullet P_X(3) &= P(X=3) = \left(\frac{1}{2}\right) \cdot \left(\frac{1}{2}\right) \cdot \left(\frac{1}{2}\right) \\ &= \overset{T}{\left(\frac{1}{2}\right)^2} \cdot \overset{T}{\left(\frac{1}{2}\right)} \quad H \end{aligned}$$

$$\begin{aligned} \bullet P_X(x) &= P(X=x) \\ &= \left(\frac{1}{2}\right)^{x-1} \cdot \left(\frac{1}{2}\right) \\ &\quad \underbrace{T \dots T}_{(x-1)} \quad H \\ &\quad \quad \quad x\text{-th} \end{aligned}$$

$$x \in \{1, 2, 3, \dots\}$$

$$\bullet P(X \leq 3) = P(\underbrace{\{X=1\} \cup \{X=2\} \cup \{X=3\}}_{\text{disjoint}})$$

$$= P(X=1) + P(X=2) + P(X=3)$$

$$= \frac{1}{2} + \left(\frac{1}{2}\right)^2 + \left(\frac{1}{2}\right)^3$$

Def. The cumulative distribution of a RV X is given by

$$F_X(x) = P(X \leq x) = \sum_{\forall k \leq x} P_X(k)$$

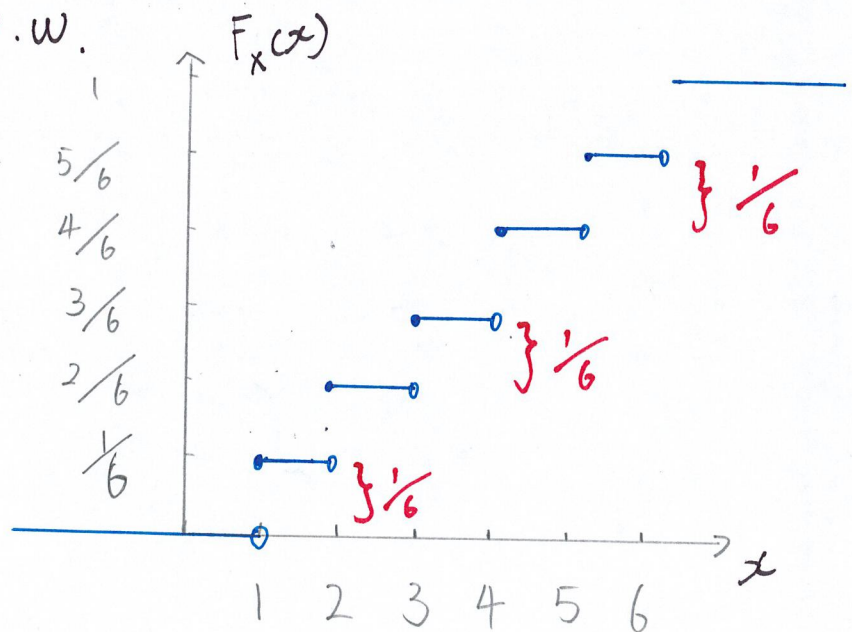
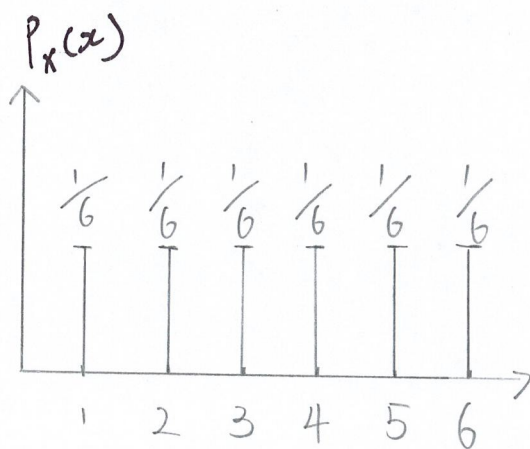
Ex. Roll a die, and let X be the result.

Find the CDF of X .

Two steps : ① Find and draw the PMF

② Draw the CDF.

$$P_X(x) = \begin{cases} \frac{1}{6} & , x = 1, 2, \dots, 6 \\ 0 & , \text{o.w.} \end{cases}$$



$$F_X(x) = \begin{cases} 0 & , x < 1 \\ \frac{1}{6} & , 1 \leq x < 2 \\ \frac{2}{6} & , 2 \leq x < 3 \\ \vdots & \\ 1 & , x \geq 6 \end{cases}$$

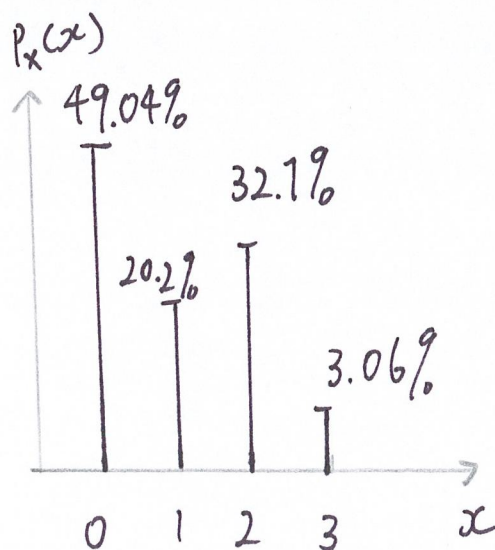
Ex. Let $\text{supp}(X) = \{0, 1, 2, 3\}$,

$$P_X(1) = 20.2\% \quad , \quad P_X(2) = 32.7\%$$

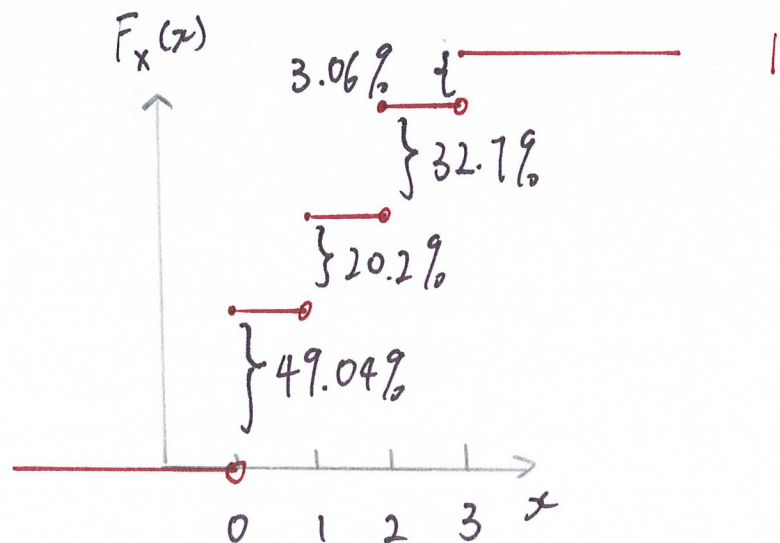
$$P_X(3) = 3.06\%$$

Find $F_X(x)$.

$$\begin{aligned} P_X(0) &= 1 - P_X(1) - P_X(2) - P_X(3) \\ &= 49.04\% \end{aligned}$$



$\Rightarrow$



$$F_X(x) = \begin{cases} 0 & , \quad x < 0 \\ 0.4904 & , \quad 0 \leq x < 1 \\ 0.6424 & , \quad 1 \leq x < 2 \\ 0.9649 & , \quad 2 \leq x < 3 \\ 1 & , \quad x \geq 3 \end{cases}$$

Properties of CDF :

- $F_X(x) = P(X \leq x) \in [0, 1]$

- $F_X(x)$ must be non-decreasing ↗ flat
↘ increasing

- $\lim_{x \rightarrow \infty} F_X(x) = P(X \leq \infty) = 1$

- $\lim_{x \rightarrow -\infty} F_X(x) = P(X \leq -\infty) = 0$